

Bridging the Knowledge Gap: The Influence of Strong Ties, Network Cohesion, and Network Range on the Transfer of Knowledge Between Organizational Units

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Prior research has emphasized the importance of boundary spanners in facilitating the transfer of knowledge between organizational units. The successful transfer of knowledge between organizational units is critical for a number of organizational processes and performance outcomes. The empirical evidence on the success of boundary spanners is mixed, however. Research findings indicate boundary spanners can either facilitate or inhibit the flow of knowledge between organizational units. We develop and test a theoretical argument emphasizing the importance of the broader network context in which boundary spanning occurs. In particular, we consider how tie strength, network cohesion, and network range affect the level of knowledge acquired in cross-unit knowledge transfer relationships. An analysis of knowledge transfer relationships among several hundred scientists indicates that each network feature had a positive effect on the level of knowledge acquired in cross-unit knowledge transfer relationships. Our findings illustrate how network features contribute to the flow of knowledge between organizational units and, therefore, how network context contributes to heterogeneity in boundary-spanning outcomes.

Key words: social networks; organizational boundaries; knowledge management

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Introduction

The ability to transfer knowledge between organizational units is critical for a host of organizational processes and performance outcomes, including creativity, innovation, and new product development (Argote 1999, Carlile 2004, Bechky 2003, Hargadon 2003). Organizational units often represent distinct stocks of knowledge and expertise. Consequently, acquiring knowledge from outside the focal unit provides unit members with an opportunity to see a problem or task from an alternative perspective, which can stimulate creativity and innovation (Leonard-Burton 1995, Hargadon and Sutton 1997, Perry-Smith 2006, Hansen 1999). Although cross-unit transfers can be beneficial, insufficient knowledge overlap between the knowledge source and recipient makes cross-unit transfers more difficult than within-unit transfers. Cross-unit transfers can be even more problematic in a knowledge- or technology-based organization, where unit members often develop a “local” language, which makes it easier for them to learn from each other but also makes it more difficult for unit members to share knowledge with colleagues outside their unit (Dougherty 1992). Indeed, in knowledge- and technology-based organizations, it is

generally understood that cross-unit transfers are simultaneously a source and a barrier to innovation and new product development (Carlile 2002, p. 442; Seidel and O’Mahony 2010).

Prior research has highlighted the importance of individuals who maintain relationships with colleagues in different organizational units (i.e., “boundary spanners”) in facilitating cross-unit transfers (Allen and Cohen 1969). Yet the empirical evidence on the success of boundary-spanning outcomes is mixed. On the one hand, units with boundary spanners are more productive (Tushman and Katz 1980, Ancona and Caldwell 1992, Hansen 1999), and individuals that maintain a boundary-spanning position do better in terms of their technological and knowledge proficiency (Tushman and Scanlan 1981). On the other hand, boundary spanners can restrict the flow of knowledge throughout organizations (Gould and Fernandez 1989, Cross et al. 2002), sometimes because they desire to maintain their power and influence (Crozier 1964) or because they are unwilling to invest the time and effort that is required for successful knowledge transfer.

The mixed empirical evidence suggests that whatever benefits cross-unit transfers can create often go

unrealized, even when boundary spanners are present. We attempt to shed light on the drivers of variation in boundary spanners' effectiveness by recognizing that a boundary spanner often maintains more than one cross-unit transfer, and each transfer can occur in a different network context.¹ We emphasize the importance of network context because prior research has established the importance of network relationships in facilitating knowledge transfer (Szulanski 1996; Uzzi 1997; Hansen 1999; Reagans and McEvily 2003, 2008; Sgourev and Zuckerman 2011). In particular, three network features—tie strength, network cohesion, and network range—have a positive effect on the successful transfer of knowledge. The strength of the interpersonal relationship between two individuals influences their willingness to engage in knowledge transfer and the amount of effort they dedicate to the activity. The two individuals involved in the transfer could be connected by a strong network connection either directly or indirectly through mutual third-party ties (i.e., high network cohesion). Moreover, individuals who occupy a network position characterized by range develop a greater capacity for knowledge transfer, thereby increasing the likelihood of successful knowledge transfer.

Whereas we know a great deal about how the three network features contribute to knowledge transfer *in general*, we know very little about how the features contribute to cross-unit transfers *in particular*. Instead of estimating network effects directly, scholars have inferred the “network effect” on cross-unit transfers from other processes. For example, in an analysis of innovation, Tortoriello and Krackhardt (2010) found that individuals were more likely to demonstrate higher innovative performance when they maintained cross-unit knowledge transfer relationships that were characterized by closure (i.e., network cohesion). Tortoriello and Krackhardt, however, did not analyze the flow of knowledge directly. Instead, they assumed that network cohesion increased the amount of knowledge acquired during cross-unit transfers, and the successful acquisition of knowledge in cross-unit transfers contributed to innovation. Reagans and McEvily (2003, 2008) provided another example. These researchers considered how the three network features contributed to knowledge flows, but they did not distinguish within-unit transfers from cross-unit transfers in their empirical analysis.

Our research objective is to understand the extent to which tie strength, network cohesion, and network range contribute to the level of knowledge acquired in cross-unit transfers. If tie strength, network cohesion, and network range facilitate the acquisition of knowledge in cross-unit transfers, then a focus on network context could help to explain heterogeneity in boundary-spanning outcomes. For example, if strong ties facilitate the level of knowledge acquired in cross-unit transfers, and if individuals can only maintain a small number

of strong ties, we should expect to see heterogeneity in cross-unit transfers, and therefore heterogeneity in boundary-spanning outcomes, whenever an individual has developed a relatively large number of cross-unit knowledge transfer relationships.

To gain a better understanding of how tie strength, network cohesion, and network range influence cross-unit transfers, we studied the research and development division of a large multinational high-technology company. The division's mission was to establish the company as a market leader for the next 10 years. Individuals in the division were assigned to 49 distinct teams, 16 laboratories, and 21 technological road maps. The laboratories and road maps were associated with specific technologies and end-market products, and work was accomplished in the context of teams. The successful transfer of knowledge within and between teams, laboratories, and technological road maps was seen as critical by top executives for the division to accomplish its goals. Work teams, laboratories, and technological road maps represented organizational units that were characterized by specialized knowledge and expertise. We collected network and knowledge transfer data from 276 scientists in the research and development (R&D) division, and we analyzed the extent to which tie strength, network cohesion, and network range affected a scientist's ability to acquire knowledge from individuals who worked in other teams, laboratories, and technological road maps. The empirical results indicated that the level of knowledge acquired in cross-unit transfers was lower than the level of knowledge acquired in within-unit transfers, but each network feature increased the amount of knowledge acquired in cross-unit knowledge transfer relationships.

Theory and Hypotheses

Cross-Unit Knowledge Acquisition

A potential knowledge recipient is more likely to successfully acquire knowledge from a source when the source and recipient overlap in what they know. Knowledge overlap facilitates knowledge acquisition because individuals learn new ideas by associating those ideas with what they already know (Cohen and Levinthal 1990, Simon 1991). Knowledge acquisition is often defined in terms of the successful transfer of knowledge from a source to a recipient (Reagans and McEvily 2003, 2008). The acquisition process ends once a recipient (sufficiently) understands whatever knowledge the source has shared. Knowledge acquisition can also be defined in terms of “knowledge in use.” When defined in this way, achieving understanding is the initial step in a transfer process that ends when the recipient is able to use the shared knowledge in his or her area of expertise or domain.² We expect the acquisition of knowledge to be more successful when the source and recipient work in the same organizational unit because individuals

who work in the same unit are more likely to overlap in their knowledge and expertise (Cohen and Levinthal 1990). The overlap in knowledge and expertise should make it easier for the source to explain what he knows to the recipient and for the recipient to assimilate and use whatever knowledge he has acquired. When a source and recipient work in different units, the source should find it more difficult to share what he knows and the recipient should struggle to apply whatever knowledge he has acquired (Cummings and Teng 2003). The process of sharing knowledge between organizational units should be even more problematic in a knowledge- or technology-based organization because in these kinds of organizations, the source and the recipient are more likely to have developed distinct frameworks, heuristics, and coding schemes (Dougherty 1992, Bechky 2003). As a result, the source and recipient are more likely to be accustomed to communicating with their respective colleagues using more “tacit” knowledge, which should make cross-unit transfer even more difficult. With respect to knowledge acquisition, the foregoing arguments suggest that individuals on the receiving end of knowledge transfers are likely to struggle to understand and apply knowledge originating from outside their unit. Thus, we expect cross-unit knowledge acquisition to be more difficult and less successful than knowledge acquisition within units. Accordingly, we hypothesize the following association.

HYPOTHESIS 1. *Knowledge acquisition is lower in cross-unit (as opposed to within-unit) transfers.*

The first hypothesis provides a baseline for the network predictions we develop next. In particular, we focus on how the broader network context defined around cross-unit relationships can affect the baseline association between knowledge acquisition and cross-unit transfers.

Network Context and Cross-Unit Knowledge Acquisition

Sharing knowledge between units can be more difficult than sharing it within units, but the strength of the interpersonal relationship between a knowledge source and potential recipient should facilitate knowledge acquisition by increasing the likelihood that two individuals will engage in knowledge transfer and by increasing the amount of effort they are willing to exert during the transfer process (Szulanski 1996, Uzzi 1997, Hansen 1999). In general, several factors reduce the likelihood that a knowledge source will share what he knows. For example, although sharing knowledge between units can be beneficial for the recipient and the larger organization, it is costly for the source, at least in terms of the time and effort the source must allocate to the transfer process. Moreover, individuals who share what they know become more redundant with their colleagues,

which can undermine their worth and value. Tie strength can help to overcome this initial barrier to knowledge transfer because strong interpersonal connections are characterized by a greater degree of trust and cooperation (Granovetter 1973). If the source and recipient are connected by a strong interpersonal connection, the source will have fewer concerns about his knowledge being appropriated or misused (Krackhardt 1999, McEvily et al. 2003). Cross-unit transfers are less likely to succeed because of the lack of common knowledge. Strong ties are characterized by a greater degree of commitment and emotional attachment, and as a result, if the source and the recipient are connected by a strong relationship, the knowledge source is willing to spend more time and effort sharing what he knows with a potential recipient.

Thus far, we have focused on the “strong tie effect” from the perspective of the knowledge source, but a strong tie can also mitigate reservations by the recipient. In particular, requesting knowledge (or assistance in general) is risky in the sense that the individual seeking knowledge necessarily reveals his ignorance, which could undermine his standing in the organization. Moreover, in technical organizations, individuals might be skeptical of ideas and innovations developed outside their organizational unit—a phenomenon that Katz and Allen (1982) called the “not invented here syndrome.” To the extent that individuals are inclined to reject ideas and innovations developed outside their organizational unit, they are likely to avoid seeking knowledge from other units in the first place. If the recipient has a strong tie with a source working in a different unit, however, the recipient is more likely to admit his ignorance and is less likely to reject external knowledge outright, even if its value is not immediately obvious (Kane et al. 2005). Thus a strong tie increases a recipient’s willingness to initiate the knowledge acquisition process, and this willingness helps to overcome a tendency to avoid acquiring knowledge from individuals outside the focal unit. Even after a recipient has received knowledge from an external source, he must still figure out how to use it. The recipient must appraise, adapt, and ultimately “transform” whatever he has learned from the source in order for it to be useful in his work context. When the recipient has a strong tie with the source, he is more likely to willingly exert the time and effort that is often required to put the knowledge he has received to good use.

The discussion above suggests that an increase in tie strength increases the likelihood of acquiring knowledge in a cross-unit relationship because the presence of a strong network connection between two individuals increases the likelihood of them engaging in the transfer and also increases the amount of time and effort the individuals are willing to commit to the successful transfer of knowledge. Following this rationale, we expect

for the presence of a strong tie between two individuals to offset the negative association between cross-unit transfers and successful knowledge acquisition.

HYPOTHESIS 2. *The negative association between cross-unit transfers and knowledge acquisition becomes less negative as tie strength increases.*

Even if a source and recipient are not connected by a strong connection directly, they could be connected by a strong tie indirectly through the ties they have with mutual third-party contacts. In particular, when individuals are embedded in a dense web of strong third-party ties, network cohesion is high; when cohesion is high, the same set of dynamics that we described above for a strong interpersonal connection is more likely to emerge. In particular, trust, cooperation, and the desire or motivation to assist one another are more likely to develop in a cohesive or closed group (Granovetter 1985, Coleman 1988). The more prosocial behavior could result from a desire to protect one's reputation (Coleman 1990). Or, individuals could cooperate in dense network structures more frequently because individual behavior in these kinds of situations is guided by norms that define what is considered to be appropriate and inappropriate behavior (Gargiulo et al. 2009, Portes and Sensenbrenner 1993, Uzzi 1997). Whatever the source, scholars agree that prosocial behaviors occur more frequently within cohesive groups.

If two individuals are connected by a strong relationship indirectly through mutual third-party ties, they are more likely to engage in knowledge transfer and to dedicate more time and effort to the knowledge acquisition process. Based on this argument, we expect for the presence of a strong indirect tie to offset the negative association between cross-unit transfers and successful knowledge acquisition.

HYPOTHESIS 3. *The negative association between cross-unit transfers and knowledge acquisition becomes less negative as network cohesion increases.*

In addition to the effects of strong ties and network cohesion on knowledge transfer, previous research has established the importance of network range (Reagans and McEvily 2003, 2008). Individuals with more range across their personal networks maintain relationships that cut across nonredundant network clusters. Clusters in a network often represent distinct stocks of knowledge, information, and frames of reference. Therefore, maintaining a wide-ranging network is more difficult because it forces an individual to deal with more divergent knowledge and expertise and potentially conflicting ways of seeing the world. However, extensive interactions with colleagues who work in disparate areas of expertise increase the odds that an individual will become more accustomed to translating and transforming knowledge so that it can be understood and applied

in new domains. Such a transformation is an essential part of the knowledge acquisition process (Dougherty 1992, Carlile 2002, Bechky 2003). Thus, individuals occupying network positions that cut across nonredundant stocks of knowledge and expertise should develop a greater capacity for knowledge acquisition, and this capability should improve their success in cross-unit knowledge transfer relationships. This line of argument leads to our final prediction.

HYPOTHESIS 4. *The negative association between cross-unit transfers and knowledge acquisition becomes less negative as network range increases.*

Methods

The setting for our study was the R&D division of a large multinational technology company. Tortoriello and Krackhardt (2010) use data from the same firm. However, instead of focusing on the determinants of knowledge acquisition in cross-unit transfers, they concentrated on the implications of maintaining embedded cross-unit transfers for innovation. At the time of our survey, the R&D division consisted of 16 geographically dispersed laboratories. Four of the laboratories were located in the United States, 10 in Europe, and 2 in Asia. The laboratories produced a variety of products, including silicon-based microelectronic components such as memories, transistors, wireless applications, integrated circuits amplifiers, and microprocessor peripherals. Targeted markets included digital consumer electronics, wireless communications, storage and security systems, as well as a broad range of applications for specific market segments such as aerospace, automotive, advanced computing, imaging, and audio systems. A total of 276 individuals worked across the 16 laboratories, and those individuals were assigned to 1 of 21 technological road maps. Individuals who were assigned to a technological road map were responsible for achieving milestones in the development of a specific technology. Bluetooth, computer architecture, connectivity, imaging, networking, low-power devices, optics, radio, rendering, security, and terminals were examples of technological road maps. Technological road maps were associated with a corresponding family of actual products. For example, connectivity dealt with wireless data transmissions. Routers, smartphones, and portable computers used wireless data transmissions. Another example was imaging. Individuals assigned to imaging developed and identified algorithms to optimize image visualization, and imaging was relevant for a number of devices, including handhelds, smartphones, and computer screens. The technological road maps often cut across laboratories such that individuals who worked in the same laboratory were often assigned to different technological road maps, and individuals who were

assigned to the same road map often worked in different laboratories. Individuals in the R&D unit were also assigned to 49 different teams; the teams were important because research in the division was frequently carried out in such groups.

We collected data from multiple sources. First, after extensive field interviews with the company's senior managers and several researchers from different laboratories and road maps, we developed a questionnaire to measure knowledge sharing relationships that existed between individual members of the division. Using a semistructured protocol, we interviewed 24 people: 4 senior managers and 20 researchers. Of the researchers, there were eight senior, six mid-career, and six junior researchers. Following the interviews, we then pretested the Web-based questionnaire on 12 individuals who did not take part in the study.

The knowledge sharing data were collected using a sociometric approach. Respondents were first presented with a fixed roster that listed the names of all 276 researchers in the division. Individuals were instructed to identify the names of "colleagues who represented an important source of technical knowledge in the past two years." Once respondents identified their contacts, they were then asked to describe their relationships with each cited contact in terms of communication frequency and knowledge acquisition. A total of 249 individuals (or about 91% of the study population) completed the entire questionnaire. On average, respondents identified approximately 11 (10.72) contacts.³ We supplemented our questionnaire data with archival data provided by the company. The archival data allowed us to measure a number of individual characteristics, including each respondent's assigned team, laboratory, technological road map, job grade, education level, organizational tenure, gender, and laboratory leadership.⁴

Our network data collection strategy, although appropriate for the objectives of the study, relies on two important assumptions. In particular, we assumed that scientists in the R&D division were the most relevant knowledge sources, and that knowledge flowed across knowledge transfer relationships and not other kinds of network connections. These two assumptions define the boundary of our network (Laumann et al. 1983, p. 18). These assumptions are important because when the boundary specification choices exclude important actors and/or connections, measurement error is introduced. Our data collection strategy was limited in (at least) two important ways. First, in addition to asking respondents to indicate knowledge sources, respondents should have been allowed to nominate knowledge recipients. Second, respondents should have been given an opportunity to name external contacts (i.e., contacts outside of the division or outside the focal organization) as possible sources and/or recipients of knowledge. Because respondents could not name external contacts and were only asked to identify knowledge sources, our

network data were measured with error. We do not know how much measurement error was introduced into our network data because we do not know who the relevant actors were and how those actors were connected to each other.

An alternative approach to collecting our network data would have been to use a modified version of the strategy used by Reagans and McEvily (2008). We could have supplemented the R&D roster with name-generator questions to elicit the names of external contacts. External contacts would then be added to the R&D roster, and we could have asked respondents to indicate how frequently they communicated with each individual on the list. We would focus on communication frequency because individuals recall the frequency of their interactions with considerable accuracy (Reason 2008, p. 24). Next, we could have let respondents nominate individuals who were important knowledge sources and/or recipients, and once we had identified the relevant set of knowledge transfer relationships, we could have asked questions about the level of knowledge acquired in each relationship. Although introducing external contacts would allow us to collect more network data, even this strategy does not eliminate entirely the potential for measurement error. How much measurement error is introduced by different data collection strategies is an important, but relatively understudied, question (Laumann et al. 1983). We now have the means to make progress on this important question. For example, improvements in information technology have allowed us to collect network data through less intrusive means, including e-mail communication and electronic name tags that measure the frequency of face-to-face interactions (Wu et al. 2011). Although these techniques do not necessarily reveal the "true" network structure, they would at least provide estimates of actors and relationships that occur most frequently. Data collection techniques that miss these actors and relationships are likely to introduce the most measurement error. Although we are unable to address this question in the current study, we believe it is an important area for future inquiry.

Variables

Dependent Variable

Knowledge Acquisition. Successful knowledge acquisition was our dependent variable. Extensive field interviews with senior managers and scientists indicated that members of the R&D division described successful knowledge transfer in terms of acquisition. This suggests that for cross-unit transfers to be successful, the knowledge acquired from a contact located in a different organizational unit had to be recognized, understood, and adapted in a way that made it useful for the recipient. Accordingly, in our study we focused on *knowledge acquisition* as the outcome of interest, examined

from the point of view of the knowledge recipient. Interviewees emphasized that they viewed knowledge transfer as truly successful when two individuals were able to *adapt* and *use* whatever knowledge they shared with each other. For example, one engineer we interviewed stated,

Cooperation between engineers and mathematicians is critical to finding solutions that are stable and that can be implemented. At the same time, it is also important to explain your ideas in terms that people in production can understand. I can't use the same language I use with mathematicians there. Constant interactions help in the adaptation of knowledge.

The interviews also indicated that firm members thought successful knowledge transfer was more likely if the two individuals first recognized where their knowledge and expertise overlapped (i.e., recognize similarities) but were also able to recognize where their knowledge and expertise did not overlap (i.e., understand differences). Along these lines, one researcher with whom we spoke noted,

I always try to explain a problem differently depending on who I am interacting with. The process is efficient only when we [me and my counterpart] are able to transform the knowledge we are sharing.

The four items in Table 1 capture different aspects of successful knowledge acquisition in the division. For each contact in their network, respondents evaluated the four items reported in Table 1 on a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). Factor analysis revealed that the items loaded on a single factor. The first principal component explained 69% of the variance, and loadings on the first principle component ranged from 0.78 to 0.87. Cronbach's alpha for the four items was 0.85. We calculated the mean of the four items in Table 1 to define the level of successful knowledge acquisition.⁵

Independent Variables

Cross-Unit Transfer. We assumed that each team, laboratory, and technological road map represented a distinct stock of knowledge and expertise, and we defined a cross-unit transfer in terms of the number of organizational units it spanned. Specifically, if the focal knowledge transfer relationship occurred between two individuals who shared membership in all three types of organizational units (i.e., same team, same laboratory, and same technological road map), our cross-unit transfer variable was set equal to 0. If the two individuals shared membership in two units, our cross-unit variable was set equal to 1, and if they shared membership in only one unit, our cross-unit transfer variable was set equal to 2. If the two individuals did not share membership in any units, our cross-unit transfer variable was

Table 1 Knowledge Acquisition Items

	Mean	SD	Loading
It is easy for me to recognize the similarities between my knowledge and expertise and the knowledge and expertise of this person	3.75	0.99	0.83
It is easy for me to understand the differences between my knowledge and expertise and the knowledge and expertise of this person	3.99	0.91	0.78
It is easy for me to <i>transform</i> and <i>adapt</i> my knowledge toward the implementation of new innovative solutions when working with this person	3.84	0.95	0.85
It is easy for me to <i>effectively use</i> this person's knowledge in connection with what I already know	3.97	0.96	0.87

Note. Items were measured on a five-point Likert scale ranging from "strongly disagree" to "strongly agree."

set equal to 3. We had four types of mutually exclusive knowledge transfer relationships, and we created indicator variables for the three relationships that cut across at least one organizational unit. If the knowledge transfer relationship spanned one boundary, our *one unit* indicator variable was set equal to 1 and remained equal to 0 otherwise. If the knowledge transfer relationship spanned two boundaries, our *two units* indicator variable was set equal to 1 and remained equal to 0 otherwise. If the knowledge transfer relationship spanned all three boundaries, our *three units* indicator variable was set equal to 1 and remained equal to 0 otherwise.

Our cross-unit transfer indicators were consistent with the idea that the more organizational units a transfer spanned, the less knowledge the two individuals involved in the transfer would have in common and the more difficult it would be to share knowledge. At the same time, we recognize that our cross-unit indicators had limitations. In particular, with respect to measuring the absence of knowledge overlap, there was considerable heterogeneity within each kind of cross-unit transfer that our indicator variables did not capture. By defining cross-unit transfer in terms of the number of organizational units the relationship spanned, we assumed that each team, laboratory, and technological road map represented a distinct stock of knowledge and expertise. Thus, we treated a knowledge transfer relationship that occurred between any two teams as spanning just as much of the knowledge space as a transfer relationship that spanned any two laboratories or any two technological road maps (i.e., a two-unit transfer). We would have preferred to define a cross-unit transfer in terms of the actual amount of knowledge a source and recipient had in common. Individuals who work in different units are less likely to share knowledge and expertise

(Dougherty 1992, Bechky 2003), but using cross-unit transfers as a proxy for knowledge overlap was an important limitation. A more systematic test of our argument would have measured how tacit shared knowledge was and how much the source and recipient overlapped in what they knew.⁶

Tie Strength. Each respondent was asked to describe the strength of his or her relationship with each cited contact. Tie strength was measured in terms of communication frequency, which ranged from 1 (i.e., rarely) to 5 (i.e., very frequently).⁷ Although tie strength has been conceptualized in terms of communication frequency and emotional closeness (Burt 1992), empirically, the two dimensions of tie strength are often highly correlated (Hansen 1999, p. 94; Reagans and McEvily 2003). Following prior research, we measured tie strength in terms of the proportion of available network time and energy a respondent allocated to a focal contact (Burt 1992, Uzzi 1996).⁸

$$m_{ij} = \frac{z_{ij} + z_{ji}}{2},$$

$$p_{ij} = \frac{m_{ij}}{\sum_q m_{iq}}.$$

To calculate network proportions, we first divided every relationship in a respondent's network by 2, which was the strongest relationship a respondent could have had with any of his or her contacts. The term m_{ij} measures the marginal strength of the relationship from the respondent to a focal contact. We then summed the marginal strength relationships across every contact q to define the total amount of network time available to the focal individual. We then divided each m_{ij} by the total amount of network time available to each individual to define p_{ij} , which measures the proportion of network time and energy respondent i allocated to contact j . Larger values of p_{ij} indicate relatively stronger relationships (e.g., relatively more frequent communication), whereas smaller values indicate weaker relationships (e.g., relatively less frequent communication).

As summarized in Hypothesis 2, we expected for an increase in tie strength to offset any negative association between cross-unit transfers and successful knowledge acquisitions. We tested this prediction by including an interaction between tie strength and our three indicators of cross-unit transfers. The combined influence of the main effect for tie strength and the interaction between tie strength and the cross-unit transfers allowed us to examine the extent to which an increase in tie strength offset the negative association between cross-unit transfers and knowledge acquisition.

Network Cohesion. To capture the extent to which a knowledge sharing relationship was embedded in a dense web of third-party ties, we used Burt's measure

of indirect structural constraint (Burt 1992, pp. 54–56). Indirect constraint assigns higher scores to interactions that were surrounded by strong third-party ties. Stated formally, indirect constraint was defined as follows:

$$c_{ij} = \sum_q p_{iq} p_{qj},$$

where p_{iq} reflects the proportion of i 's network time devoted to colleague q , and p_{qj} reflects the proportion of q 's network time allocated to contact j . The term c_{ij} indicates the extent to which i and j 's relationship was surrounded by relatively strong third-party ties. When c_{ij} was high, the network surrounding i and j 's relationship was characterized by closure, or a high level of network cohesion (Reagans and McEvily 2003).

Hypothesis 3 predicted that an increase in network cohesion would offset the negative association between cross-unit transfers and knowledge acquisition. As was the case with tie strength, we tested this prediction by including an interaction between network cohesion and our three indicators of cross-unit transfers. The combined influence of the main effect for network cohesion and the interaction between cohesion and the cross-unit transfers allowed us to examine the extent to which an increase in network cohesion offset the negative association between cross-unit transfers and knowledge acquisition.

Network Range. Our indicator of network range was information centrality (Stephenson and Zelen 1989, Brandes and Fleischer 2005), which we computed using UCINET VI (Borgatti et al. 2002). Information centrality, like closeness centrality, is a function of the path distance (i.e., geodesic) between any two actors. Unlike closeness centrality, however, which gives maximum weight to the shortest path between two actors, information centrality considers all paths and assigns greater weights to shorter paths. Individuals who were high on information centrality were connected to nonequivalent contacts in the knowledge transfer network, and thus individuals who were high on information centrality were more likely to be exposed to nonredundant knowledge and information.

The fourth hypothesis predicted that an increase in network range would offset the negative association between cross-unit transfers and knowledge acquisition. Once again, we tested this prediction by including an interaction between network range and our three indicators of cross-unit transfers. The combined influence of the main effect for network range and the interaction between network range and the cross-unit transfers allowed us to examine the extent to which an increase in network range offset the negative association between cross-unit transfers and knowledge acquisition.

Control Variables

We controlled for a number of demographic characteristics that could have determined the success of the knowledge acquisition process. As noted above, we had demographic data with respect to gender, education, tenure, job grade, and laboratory leadership (i.e., the individual is a laboratory leader). These characteristics could have affected knowledge acquisition. For example, being a leader in a laboratory could have affected knowledge acquisition, but the nature of the association with leadership and acquisition was not clear. Individuals could have either been motivated to share knowledge with the laboratory leader or been reluctant to be involved in such exchanges. It was also possible that individuals felt more comfortable sharing what they knew with colleagues of the same standing in the laboratory. To control for the potential influence of these kind of status-based dynamics, we created four types of relationships with the leadership data (i.e., both leaders, source was a leader and recipient was not a leader, recipient was a leader and source was not a leader, and source was not a leader and recipient was not a leader). Relationships involving two nonleaders were the excluded category. Gender-based dynamics could have affected knowledge acquisition as well. Men could have felt more comfortable sharing knowledge with other men, and women could have felt more comfortable sharing what they know with other women. To control for the influence of gender, we created a categorical variable that varied with the gender of the source and recipient (i.e., female recipient and female source, male recipient and male source, male source and female recipient, female source and male recipient). Relationships that involved two women were the excluded category. The members of the organization also varied in terms of education level (high school diploma, B.A., M.A., or Ph.D.). Preliminary analysis indicated that a critical distinction was between individuals who graduated from high school versus those who had more formal education. To control for the influence of education similarity, we created a categorical variable that varied with the education level of the source and recipient (i.e., high school recipient and high school source, more than high school recipient and high school source, high school recipient and more than high school source, and more than high school recipient and more than high school source). Relationships that involved two individuals with more than a high school degree were the excluded category.⁹ Individuals in our sample also varied in terms of job grade and organizational tenure. To account for the influence of job grade, we controlled for the job grade of the source, the job grade of the recipient, and the product of the two job grade variables.¹⁰ To account for the influence of organizational tenure, we controlled for the source's organizational tenure, the recipient's organizational tenure, and the product of the two tenure variables

(Reagans 2011). Finally, we controlled for network size, which was defined as the number of knowledge transfer relationships that each respondent had. The sheer size of each respondent's network was a potential confounding factor in that it could have defined an individual's ability to access different pools of knowledge and his or her ability to share knowledge with his or her colleagues.

Analysis

The 249 individuals who completed the survey were involved in a total of 2,370 knowledge transfer relationships. Because there were multiple observations for each respondent and for each contact, the observed knowledge transfer relationships were not independent. This kind of clustering is a violation of the independence assumption in regression analysis and can reduce the size of our standard errors, thereby artificially increasing the significance of our tests. To adjust our standard errors for clustering, we introduced a random effect for every focal individual (i.e., knowledge recipient) and every contact (i.e., knowledge source) in our analysis. The two-way random effect is one solution to the nonindependence problem that dyad-level analyses can introduce (Krackhardt 1987, 1988; Simpson 2001). The two-way individual random effects can also control for unmeasured characteristics of a recipient and/or source that could have affected the outcome of the knowledge acquisition process.

We focus on successful knowledge acquisition in our empirical analysis, but the knowledge transfer relationships we observed were a subset of the connections that could have been observed. This kind of network endogeneity could bias our parameter estimates. To adjust our estimates for network endogeneity, we identified a set of control variables that affected the likelihood that a recipient would receive knowledge from a source but did not affect our knowledge acquisition variable (Azoulay et al. 2011). Empirical analysis indicated that our lab leadership and seniority variables affected the likelihood that a recipient would receive knowledge from a source but did not have an effect on knowledge acquisition. Using probit regression, we predicted the likelihood of a recipient receiving knowledge from a source as a function of our lab leadership and seniority variables. The estimates indicated the lab leaders were more likely to seek knowledge and lab leaders were especially likely to seek knowledge from each other. Senior colleagues were sought out more frequently, but there was also significant evidence for homophily with respect to seniority. More junior and more senior individuals were more likely to share knowledge with individuals who were similar in seniority. The predicted "probability of sharing knowledge" from the probit regression was transformed (i.e., inverse Mills ratio) and introduced as an additional control variable in our knowledge acquisition equations.

Although the predicted probability of sharing knowledge allows us to control for one kind of unobserved heterogeneity, it does not rule out the issue all together. In particular, it is possible that any network effect we observe will simply reflect unmeasured features of individuals that affect their ability to build relationships and to successfully acquire knowledge. For example, it is possible that an individual's level of absorptive capacity (Cohen and Levinthal 1990) affected the strength of his or her knowledge transfer relationships and his or her position in the broader knowledge transfer network. Individuals with more absorptive capacity would be more likely to acquire knowledge effectively. Although the influence of unobserved individual heterogeneity cannot be ruled out all together, we took a number of steps to limit its influence. First, we controlled for the likelihood of an individual being involved in a specific knowledge transfer relationship. Second, we controlled for a number of factors that should have been correlated with an individual's ability to share knowledge, including network size, job grade, and education level. Finally, our two-way random-effects model allowed us to control for unmeasured features of the two individuals involved in a knowledge transfer relationship that could have affected knowledge acquisition, and those unmeasured features were assumed to be *uncorrelated* with the control variables and the other predictors in our model. Although we believe the controls we have introduced minimize the influence of unobserved individual heterogeneity, we also believe that a more systematic treatment of the issues we have discussed would consider the network effects in a dynamic framework (Ibarra et al. 2005), including analyzing how individual characteristics, such as absorptive capacity, affect the network formation and knowledge acquisition processes. It is possible that sharing knowledge effectively creates strong network connections, but because we do not have network and knowledge acquisition data over time, we cannot examine this issue directly.

Results

Descriptive statistics and the correlation matrix are reported in Table 2. The results of the regression models are reported in Table 3. We entered our predictions across Table 3 in a stepwise fashion. The cross-unit indicators and the control variables were introduced first. We then entered our network variables to illustrate how much each network feature contributed to successful knowledge acquisition by itself and after controlling for the influence of the other two network features. In Models 6–8, for each network feature we included the interactions with the focal network feature and our three indicators of cross-unit transfers. Model 9 includes all of the network variables and all of the interactions terms. We focus on the results in Model 9 in our discussion.

Given the large number of interactions in our model, we estimated “simple slopes” using the MARGINS command in STATA. With respect to our first prediction, we expected for knowledge acquisition to be lower in cross-unit transfers. The empirical results provide support for Hypothesis 1. The coefficients indicate that when compared to within-unit transfers, the level of knowledge acquired in one-unit ($b = -0.08$, $p < 0.001$), two-unit ($b = -0.11$, $p < 0.001$), and three-unit ($b = -0.15$, $p < 0.001$) knowledge transfer relationships was lower. The coefficients also indicate that each network feature had a positive effect on knowledge acquisition in general. An increase in tie strength ($b = 2.9$, $p < 0.001$), network cohesion ($b = 1.6$, $p < 0.001$), and network range ($b = 0.40$, $p < 0.001$) increased the level of knowledge acquired. The average network effects were consistent with prior theoretical arguments that emphasize the importance of the three network features for knowledge acquisition (Reagans and McEvily 2003, 2008).

The coefficients discussed above show the average network effects. Our three network predictions focused on how each network feature affected the level of knowledge acquired in cross-unit transfers. Given that the main effect for each network variable was positive and significant, for us to reject a prediction, the interaction between a kind of cross-unit transfer and the focal network variable (e.g., the interaction between tie strength and one-unit transfers) would have to be negative *and* large enough to offset the magnitude of the positive main effect. The network effect for each kind of knowledge transfer relationship is illustrated in Table 4. Each row is a kind of knowledge transfer relationship (e.g., one unit) and each column is a network feature (e.g., tie strength). Each cell entry is the effect of an increase in the network variable (column) on the acquisition of knowledge in a kind of transfer relationship (row). The coefficients in Table 4 provide strong support for our second, third, and fourth predictions. An increase in any network feature had a positive effect on the level of knowledge acquired in a cross-unit transfer. There is, however, one finding in Table 4 that is inconsistent with prior research. Network cohesion did not have an effect on the level of knowledge acquired in within-unit transfers. Although we cannot explain why network cohesion did not have an effect on the level of knowledge acquired in within-unit transfers, the estimated effect validates our framework by illustrating why it is problematic to assume a particular network effect is constant across different kinds of knowledge transfer relationships.

Network Mechanisms

The coefficients in Table 4 illustrate how tie strength, network cohesion, and network range contributed to the level of knowledge acquired in different kinds of knowledge transfer relationships. The results also illustrate why we observe heterogeneity in boundary-spanning

Table 2 Descriptive Statistics and Correlations

Variables	Mean	SD	1	2	3	4	5	6	7	8
1 Knowledge acquisition	3.94	0.79								
2 Probability of sharing knowledge	0.78	0.01	0.070							
3 Recipient female–Source male	0.09	0.28	0.028	0.054						
4 Recipient male–Source Female	0.07	0.25	–0.038	0.039	–0.084					
5 Recipient male–Source male	0.83	0.37	0.004	–0.075	–0.686	–0.610				
6 Recipient greater than HS–Source HS	0.03	0.16	–0.037	–0.106	–0.011	–0.044	0.044			
7 Recipient HS–Source greater than HS	0.02	0.13	–0.050	–0.180	–0.040	0.029	0.015	–0.021		
8 Recipient HS–Source HS	0.00	0.03	0.006	–0.109	–0.009	–0.008	0.013	–0.005	–0.004	
9 Recipient's job grade	14.05	1.99	–0.008	–0.285	–0.094	–0.041	0.110	0.034	0.168	0.021
10 Source's job grade	14.44	2.10	–0.115	–0.208	–0.008	–0.066	0.064	0.144	0.051	0.022
11 R's job grade × S's job grade	53.79	24.79	–0.065	–0.254	–0.015	–0.015	0.022	0.002	0.095	0.011
12 Network size	19.97	9.18	–0.017	–0.116	–0.074	–0.041	0.086	0.044	0.052	0.003
13 One unit	0.24	0.42	0.098	–0.001	–0.012	–0.023	0.020	–0.045	–0.019	–0.016
14 Two units	0.23	0.42	–0.024	0.034	0.007	–0.042	0.021	0.076	0.005	–0.016
15 Three units	0.23	0.42	–0.206	–0.099	–0.019	–0.010	0.023	0.037	0.037	0.045
16 Tie strength	0.07	0.04	0.214	0.090	0.014	–0.001	–0.013	–0.057	–0.056	–0.022
17 Network cohesion	0.03	0.02	0.209	0.123	0.012	0.011	–0.007	–0.070	–0.076	–0.026
18 Network range	2.67	0.40	0.030	–0.105	–0.117	–0.034	0.120	0.035	0.068	0.008
Variables	9	10	11	12	13	14	15	16	17	
9 Recipient's job grade										
10 Source's job grade	0.323									
11 R's job grade × S's job grade	0.325	0.115								
12 Network size	0.607	0.194	0.208							
13 One unit	–0.030	–0.022	–0.113	–0.050						
14 Two units	0.021	0.093	–0.036	0.026	–0.307					
15 Three units	0.262	0.229	0.108	0.278	–0.360	–0.358				
16 Tie strength	–0.359	–0.218	–0.067	–0.581	0.078	–0.064	–0.307			
17 Network cohesion	–0.387	–0.209	–0.122	–0.468	0.187	–0.115	–0.436	0.606		
18 Network range	0.557	0.180	0.177	0.881	–0.039	0.064	0.277	–0.644	–0.516	

Notes. Correlations greater than |0.04| are significant at $p < 0.05$. HS, high school; R, recipient; S, source.

outcomes. For example, if a source and recipient were not connected by a strong interpersonal connection, either directly or indirectly through mutual strong third parties, the level of knowledge acquired in a cross-unit transfer was lower. Strong interpersonal connections increased the amount of knowledge acquired in cross-unit transfers, but because individuals have a limited amount of network time and energy, the results for tie strength and network cohesion also indicated that successfully acquiring knowledge in one transfer often came at the expense of acquiring knowledge in nonfocal relationships. In addition to advancing our understanding of why we see heterogeneity in boundary-spanning outcomes, the results have implications for social capital research defined more broadly. In particular, the coefficients in Table 4 can be used to provide indirect tests of two network mechanisms that are often used to develop theoretical arguments and predictions but that are difficult to observe and test directly—the network mobilization and network capabilities arguments. Several network theories maintain that strong ties represent a resource (i.e., social capital) that individuals can mobilize to achieve some desired outcome (Coleman 1988, Gould 1991, Smith 2005, Smith et al. 2012). The most

direct test of the mobilization argument would be to measure the resources activated in a relationship and the effect that tie strength and/or network cohesion has on the mobilization process. However, it is often difficult to know a priori which resources will be mobilized and when those resources will be mobilized. It is possible, however, to assess the relative ease or difficulty of achieving a desired outcome and thereby indirectly determine the value of mobilizing a relationship. Faced with a more difficult task, individuals should activate whatever resources a strong tie represents more frequently or with more intensity in order to accomplish the task; consequently, the observed value of a strong tie should more closely approximate its maximum value. The empirical results indicate that acquiring knowledge in cross-unit transfers was more difficult than acquiring knowledge in within-unit transfers. If the network mobilization argument is correct, whatever positive effect strong ties had on knowledge acquisition in general should have been even larger in cross-unit transfers. If we compare the magnitude of the strong tie effect for within-unit and three-unit transfers, the magnitude of the tie strength effect was larger in three-unit versus within-unit transfers ($\chi^2 = 6.06$, $p < 0.05$), and the magnitude

Table 3 Effect of Network Context on Knowledge Acquisition Between Units

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Controls									
<i>Probability of sharing knowledge</i>	0.546 (0.573)	0.306 (0.552)	0.398 (0.566)	0.574 (0.573)	0.302 (0.549)	0.236 (0.549)	0.353 (0.563)	0.563 (0.572)	0.204 (0.546)
<i>Recipient female–Source male</i>	−0.013 (0.058)	0.003 (0.056)	−0.017 (0.057)	−0.014 (0.058)	0.000 (0.056)	0.007 (0.055)	−0.019 (0.057)	−0.017 (0.058)	−0.002 (0.055)
<i>Recipient male–Source female</i>	−0.119 (0.073)	−0.101 (0.072)	−0.126 ⁺ (0.072)	−0.133 ⁺ (0.073)	−0.136* (0.069)	−0.096 (0.071)	−0.131 ⁺ (0.072)	−0.133 ⁺ (0.073)	−0.134 ⁺ (0.069)
<i>Recipient male–Source male</i>	−0.069 (0.072)	−0.065 (0.071)	−0.081 (0.071)	−0.083 (0.071)	−0.100 (0.068)	−0.062 (0.070)	−0.089 (0.071)	−0.086 (0.071)	−0.103 (0.068)
<i>Recipient greater than HS–Source HS</i>	0.003 (0.051)	0.014 (0.052)	0.008 (0.050)	0.004 (0.051)	0.018 (0.053)	0.013 (0.051)	0.008 (0.049)	0.003 (0.051)	0.016 (0.052)
<i>Recipient HS–Source greater than HS</i>	−0.119 (0.124)	−0.083 (0.128)	−0.103 (0.124)	−0.136 (0.122)	−0.115 (0.117)	−0.095 (0.125)	−0.108 (0.123)	−0.133 (0.122)	−0.118 (0.116)
<i>Recipient HS–Source HS</i>	0.288 (0.223)	0.324 (0.218)	0.282 (0.221)	0.273 (0.222)	0.287 (0.212)	0.340 (0.216)	0.292 (0.220)	0.270 (0.222)	0.298 (0.211)
<i>Recipient's job grade</i>	0.022* (0.010)	0.014 (0.010)	0.025* (0.010)	0.023* (0.009)	0.018* (0.009)	0.015 (0.010)	0.025** (0.010)	0.023* (0.009)	0.019* (0.009)
<i>Source's job grade</i>	−0.011** (0.004)	−0.007 ⁺ (0.004)	−0.012** (0.004)	−0.011** (0.004)	−0.007 ⁺ (0.004)	−0.007 ⁺ (0.004)	−0.012** (0.004)	−0.011** (0.004)	−0.008 ⁺ (0.004)
<i>R's job grade × S's job grade</i>	−0.006*** (0.002)	−0.007*** (0.001)	−0.005*** (0.002)	−0.006*** (0.002)	−0.007*** (0.001)	−0.006*** (0.001)	−0.005** (0.002)	−0.005** (0.002)	−0.006*** (0.001)
<i>Network size</i>	0.001 (0.002)	0.011*** (0.002)	0.004 ⁺ (0.002)	−0.008* (0.004)	−0.008* (0.004)	0.011*** (0.002)	0.004 ⁺ (0.002)	−0.006 (0.004)	−0.005 (0.004)
Cross-unit transfers									
<i>One unit</i>	−0.061** (0.019)	−0.040* (0.018)	−0.051** (0.019)	−0.063*** (0.019)	−0.041* (0.018)	−0.113** (0.035)	−0.125*** (0.038)	0.202 (0.121)	0.034 (0.176)
<i>Two units</i>	−0.128*** (0.020)	−0.084*** (0.019)	−0.093*** (0.020)	−0.132*** (0.020)	−0.076*** (0.019)	−0.196*** (0.036)	−0.198*** (0.036)	0.217 (0.131)	0.108 (0.186)
<i>Three units</i>	−0.252*** (0.019)	−0.193*** (0.019)	−0.192*** (0.020)	−0.255*** (0.019)	−0.174*** (0.020)	−0.377*** (0.036)	−0.331*** (0.034)	0.159 (0.136)	0.044 (0.177)
Network variables									
<i>Tie strength (TS)</i>		2.657*** (0.189)			2.679*** (0.207)	1.749*** (0.279)			2.383*** (0.342)
<i>Network cohesion (NC)</i>			3.208*** (0.404)		1.313** (0.429)		1.210 ⁺ (0.647)		0.134 (0.694)
<i>Network range (NR)</i>				0.201** (0.072)	0.462*** (0.071)			0.269*** (0.075)	0.489*** (0.077)
Interactions									
<i>One unit × TS</i>						0.726* (0.363)			0.168 (0.466)
<i>Two units × TS</i>						1.247** (0.383)			0.315 (0.529)
<i>Three units × TS</i>						2.871*** (0.469)			1.455* (0.591)
<i>One unit × NC</i>							1.651 ⁺ (0.877)		1.486 (0.971)
<i>Two units × NC</i>							2.720** (0.900)		1.641 (1.034)
<i>Three units × NC</i>							5.661*** (1.074)		2.759* (1.161)
<i>One unit × NR</i>								−0.106* (0.047)	−0.064 (0.056)
<i>Two units × NR</i>								−0.137** (0.050)	−0.109 ⁺ (0.060)
<i>Three units × NR</i>								−0.158** (0.051)	−0.143* (0.057)
Constant	0.620 (0.455)	0.370 (0.439)	0.575 (0.449)	0.256 (0.473)	−0.480 (0.455)	0.508 (0.436)	0.703 (0.448)	0.074 (0.476)	−0.431 (0.464)
Chi-square	261.139	483.262	331.836	268.935	536.215	528.923	364.615	281.407	593.323
Log-likelihood	−411.913	−318.835	−380.784	−409.756	−296.466	−299.531	−364.025	−410.86	−276.322
Number of observations	2,370	2,370	2,370	2,370	2,370	2,370	2,370	2,370	2,370

Notes. Coefficients are from a cross-classified (source and recipient) random-effects model. Coefficients are unstandardized, and standard errors are in parentheses. HS, high school; R, recipient; S, source.

⁺ $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Table 4 Network Effects Vary with Type of Knowledge Transfer Relationship

Type of knowledge transfer relationship	Network feature		
	Tie strength	Network cohesion	Network range
Within-unit	2.38, $p < 0.001$	0.13, $p = 0.84$	0.48, $p < 0.001$
One unit	2.55, $p < 0.001$	1.62, $p < 0.05$	0.42, $p < 0.001$
Two units	2.66, $p < 0.001$	1.77, $p < 0.05$	0.38, $p < 0.001$
Three units	3.89, $p < 0.001$	2.89, $p < 0.05$	0.34, $p < 0.001$

Note. Cell entries are “marginal effects,” or the effects that each network feature had on the successful acquisition of knowledge for a specific kind of knowledge transfer relationship.

of the network cohesion effect was larger in three-unit versus within-unit transfers ($\chi^2 = 5.65$, $p < 0.05$).

Our fourth prediction was premised on the idea that occupying a network position characterized by range introduced more cosmopolitan capabilities, including the ability to share knowledge effectively (Reagans and McEvily 2003, Reagans and Zuckerman 2008). As with mobilization, the best test of this argument would be to measure the effect that a change in network range has on a change in an individual’s capabilities. Like mobilization, however, it would be difficult to know which capabilities to measure and when the change in capabilities would materialize. If network range had a positive effect on knowledge acquisition because the individuals occupying network positions characterized by range acquire an enhanced ability to transfer knowledge, whatever positive network range effect we observed should have been smaller for cross-unit transfers. The magnitude of the ability effect should decline as the focal task increases in degree of difficulty. If we compare the magnitude of the network range effect for within-unit transfers to the magnitude of the network range effect for three-unit transfers, the network range effect in cross-unit transfers was significantly smaller ($\chi^2 = 6.22$, $p < 0.05$).

When we compared our network effects for within-unit and three-unit transfers, we found support for the network mobilization and capabilities arguments. The effects for tie strength and network cohesion were larger in the more difficult knowledge transfer relationships, and the effect for network range was smaller. The evidence was more inconclusive when we compared within-unit transfers to one- or two-unit transfers. When we compared within-unit knowledge transfer relationships to one- and two-unit transfers, the magnitude of the tie strength effect was not larger in one-unit ($\chi^2 = 0.13$, $p = 0.71$) or two-unit ($\chi^2 = 0.35$, $p = 0.55$) transfers. The magnitude of the network cohesion effect was larger in one-unit ($\chi^2 = 2.34$, $p = 0.12$) and two-unit ($\chi^2 = 2.52$, $p = 0.11$) relationships when compared to within-unit transfers, but the differences were only marginally significant.¹¹ The magnitude of the network range effect in within-unit transfers was larger than the magnitude

of the network range effect for two-unit transfers ($\chi^2 = 3.24$, $p < 0.10$),¹² but the relative size of the network effect was not smaller if the transfer only spanned one unit ($\chi^2 = 1.28$, $p = 0.25$).

Overall, the results from our indirect tests were informative but not definitive. In particular, we found support for the network mobilization and capabilities arguments when we focused on knowledge transfer relationships that spanned three units. The results from our indirect tests were more inconclusive when we compared within-unit transfers to one- and two-unit transfers. One could object to using one- and two-unit transfers to provide an indirect test because individuals who worked in the same team, laboratory, or road map could have overlapped a great deal in what they knew. One- and two-unit relationships could have been more like within-unit transfers than three-unit relationships. Indeed, finding support for the mobilization and capabilities arguments in three-unit relationships was encouraging, because individuals were very unlikely to have any knowledge in common in these relationships, and so these kinds of relationships would provide the strongest tests of the mobilization and capability arguments. Thus although the results were not definitive, at a minimum, the results suggest that using task characteristics to provide an indirect test of proposed network mechanisms could prove to be worthwhile in future research. We hasten to add, however, that although we believe indirect tests can be worthwhile, it is important to not lose sight of the ideal. A direct test of the mobilization process might help to explain why network cohesion did not have an effect on within-unit transfers. For within-unit transfers, it is possible that individuals either did not or were unable to mobilize the resources that network cohesion represented.

Summary and Discussion

In this paper, we sought to gain insight into how three network features—tie strength, network cohesion, and network range—affected the successful acquisition of knowledge across organizational boundaries. The successful acquisition of knowledge across organizational boundaries is essential for innovation and creativity, but the absence of common knowledge between the source and recipient reduces the amount of knowledge acquired during a cross-unit transfer. Prior research has established the importance of tie strength, network cohesion, and network range for knowledge acquisition in general. Our theoretical argument and empirical analysis focused on the value of the three network features for the successful acquisition of knowledge in cross-unit transfers. Our empirical results indicate that each network feature had a positive effect on the amount of knowledge acquired in cross-unit transfers.

The research findings have important implications for scholars with an interest in the transfer of

knowledge between organizational units. In particular, scholars have emphasized the importance of boundary spanners in facilitating the flow of knowledge between organizational units. Individuals who occupy a boundary-spanning position in an organization are in a position to use their network connections to increase the amount of knowledge transferred between units. The available empirical evidence, however, indicates that boundary spanners sometimes increase the amount of knowledge flowing between units but at other times restrict knowledge flows. Our results provide some insight into this heterogeneity and illustrate the importance of considering the broader network context in which each cross-unit transfer occurs. For example, we found that strong ties had a positive effect on the amount of knowledge acquired during a cross-unit transfer. Because individuals can only maintain a limited number of strong ties, our results indicate that one should expect to see variation in boundary-spanning outcomes across strong and weak ties.

Although we have focused on a single organization in our empirical analysis, we believe our results are relevant for knowledge-intensive organizations (e.g., R&D labs, professional services firms) in which a competitive advantage is derived from the successful acquisition of knowledge across distinct stocks of knowledge and expertise. Our results can be used to highlight the challenges associated with attempting to use the different network features to increase the level of knowledge acquired in cross-unit transfers. For example, if an individual (manager) knew the relative network effects we have observed, it seems natural to assume that if he or she wanted to increase the amount of knowledge acquired during cross-unit transfers, he or she would focus on building strong cross-unit knowledge transfer relationships. If we only consider three-unit transfers, the tie strength effect equaled the network cohesion effect ($\chi^2 = 0.54$, $p = 0.46$), and the tie strength effect was larger than the network range effect ($\chi^2 = 52.51$, $p < 0.001$). Although the tie strength effect and the network cohesion effects were equal, tie strength was arguably more valuable from a practical standpoint in the sense that fewer people were required for the tie strength effect to be realized. Whereas the network cohesion effect required investments from one or more third parties to be achieved, the tie strength effect only required investments from the two individuals directly involved in the relationship.

Thus, at least when compared with network cohesion and range, a focus on tie strength seems appropriate. An exclusive focus on tie strength, however, becomes less clear if we consider the observed network effects in a more dynamic context. In particular, although tie strength had a relatively large effect on the level of knowledge acquired in a cross-unit transfer, it is difficult to know a priori which knowledge transfer relationships

will be most beneficial, and the benefits that a strong tie creates take time to develop (Baum et al. 2012). Because it is difficult to know the value of sharing knowledge a priori, individuals must decide to invest before the benefits of sharing knowledge are known. An individual who focuses on tie strength could waste time investing in less rewarding knowledge transfer relationships. Investments in strong ties become even more problematic once we consider that established, strong interpersonal connections are difficult to end and thus less likely to be terminated (Sgourev and Zuckerman 2011). Thus, an individual who focuses on strong ties runs the risk of getting “trapped” in less rewarding knowledge transfer relationships (Uzzi 1997, Gargiulo and Benassi 2000, Anjos and Reagans 2012). Even if these risks can be offset, there are limits on the number of strong ties an individual can maintain.

It is useful, therefore, to contrast the tie strength effect with the network range effect. The network range effect was much smaller in magnitude than was the strong tie effect, but the range effect was not limited to a particular knowledge transfer relationship. As a result, the positive effect that network range had on the level of knowledge acquired in cross-unit transfers could be utilized across a much larger number of cross-unit knowledge transfer relationships (Reagans and McEvily 2008). Investing in range can be risky as well. In particular, there is no guarantee that the decision to invest in network connections characterized by range will be successful (Reagans and Zuckerman 2008). Moreover, in the context of cross-unit transfers, there is no guarantee the greater capacity for knowledge transfer that network range introduces will be large enough to support a sufficient level of knowledge acquisition in cross-unit transfers. Thus, although the observed network effects are informative, they do not lend themselves to easy managerial advice. Indeed, when considered in a more dynamic context, the research findings reinforce the idea that there is no one best way to organize network connections. Each network strategy is characterized by a distinct set of rewards and challenges (Reagans and Zuckerman 2008). Although we have focused on the relative network effects we observed in the R&D division, the rewards and challenges we have discussed in this section do not depend on the relative magnitudes of the observed network effects.

The empirical results we presented represent an important advancement in our understanding of how knowledge is acquired inside organizations. Bringing together formal (i.e., boundary spanning) and informal (tie strength, network cohesion, and network range) structures, this paper proposes an integrative approach to the study of social networks and knowledge sharing that contributes to the extant debate on knowledge management and organizational learning (Argote et al. 2003). Whereas prior research has focused exclusively on the importance of boundary spanners in facilitating

the flow of knowledge between organizational units, our findings suggest that the value a boundary spanner creates is in part a function of the broader network context in which the boundary spanner operates. Our findings indicate that if the goal is to promote the effective use and successful acquisition of knowledge and expertise located in different parts of the organization, promoting boundary-spanning interactions in multiunit organizations is an important first step. Developing the broader network context to support boundary-spanning outcomes is the critical second step.

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Endnotes

¹Prior research has used organizational boundaries to define different kinds of brokers. In the typology of brokers identified by Gould and Fernandez (1989, p. 93), the boundary spanners considered in this study are analogous to the "gatekeeper"-type broker. Although fully appreciating the value of this research, we have focused on how the amount of range (i.e., brokerage) in an individual's network affects his or her capacity for knowledge transfer and therefore his or her success as a boundary spanner.

²The stage in the acquisition process also determines who is in the best position to evaluate its relative success. The source is in the best position to evaluate how easy or difficult it was to share knowledge with a recipient (Reagans and McEvily 2003). However, the recipient is in the best position to evaluate the extent to which the shared knowledge has been acquired and is in use.

³The number of cited contacts was smaller than the number cited in similar studies. For example, the respondents interviewed by Reagans and McEvily (2003) cited 16 contacts on average. The smaller number of knowledge transfer partners in our study population could have been due to a number of factors, including the fact that individuals in our study worked in small laboratories (average laboratory size equaled 25 people) that were located in different geographical areas. Because the laboratories were geographically dispersed, the scientists could have had fewer opportunities to develop knowledge transfer relationships.

⁴The archival data were also used to test for sample selection bias. Empirical analysis indicated that the individuals who responded to our survey were no different from nonrespondents with respect to the individual characteristics gleaned from the archival data.

⁵In a separate analysis not reported here, we obtained the same results using factor loadings instead of the mean of items to measure successful knowledge acquisition.

⁶We also estimated models with a cross-unit knowledge transfer defined with respect to three indicator variables: cross-team, cross-laboratory, and cross-road map. The model included the main effects and all of interactions between the cross-team, cross-laboratory, and the cross-road map variables. Moreover, we interacted the main effects, two-way interactions, and three-way interactions with the three network features. The empirical results led to the same substantive conclusions as the findings discussed in the Results section. Results are available from the authors upon request.

⁷The qualitative response categories were given quantitative values using log-linear multiplicative models (Burt 1992). Relationships where a respondent did not communicate with a colleague were assigned a value of 0, and relationships where a respondent communicated with that colleague very frequently were assigned a value of 1.

⁸The rationale for transforming the original tie strength data into network proportions is based on the idea that individuals have a limited amount of network time and energy, and knowledge acquisition is more likely to be successful when an individual dedicates a significant amount of his available time to a particular transfer. Finally, although we did not assume that our network connections were reciprocated, reciprocated ties tended to be stronger on average (Krackhardt 1999).

⁹In results to be presented, we reach the same conclusions if we estimate models that control for all 16 combinations of education. We present our results using the more parsimonious specification to conserve space in our tables.

¹⁰The demographic data were virtually complete. The education data were 98.9% complete. The gender data were 98.9% complete. Tenure data were 97.8% complete. Job grade data were 98.2% complete, and the lab leadership data were 100% complete. The missing demographic data were imputed using multiple imputation techniques. Empirical results to be presented result in the same substantive conclusions, if cases with missing data were deleted.

¹¹The test results were conservative in the sense that the null hypothesis was that the two coefficients were equal (i.e., a two-sided test). Our theoretical argument, however, predicted that the network cohesion effect would be larger in one- or two-unit transfers than in within-unit transfers. Therefore, a more accurate test would be to examine the extent to which the network cohesion effect for one- or two-unit transfers was *greater than or equal* to the network cohesion effect for within-unit transfers (i.e., a one-sided test). The results from each one-sided test indicated that we could not reject the null hypothesis that the network cohesion effect in one- or two-unit transfers was greater than or equal to the network cohesion effect for within-unit transfers.

¹²If we use a more theoretically appropriate one-sided test, we cannot reject the null hypothesis that the network range effect in one-unit transfers was less than or equal to the network range effect for within-unit transfers.

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